

WORKSHEET - NAPLAN Year 9

Measurement

Perimeter, Area and Volume

Units of Measurement

Geometry

Triangles and Other Geometric Properties



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Exam Equivalent Time: 28.75 minutes (based on NAPLAN allocation of ~1.25 min/mark)

Questions

1. Measurement, NAP-26174

Kim, Bob and Liz each measure the height of the hedge in their front yards.

- Kim's hedge is 0.72 metres tall.
- Bob's hedge is 815 millimetres tall.
- Liz's hedge is 68 centimetres tall

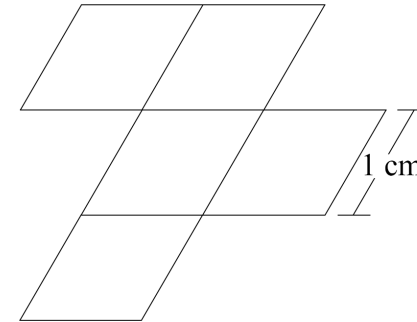
Who has the tallest hedge in their front yard?

Kim Bob Liz

☐ ☐ ☐

2. Measurement, NAP-08572

A shape, pictured below, is made with 5 rhombuses.



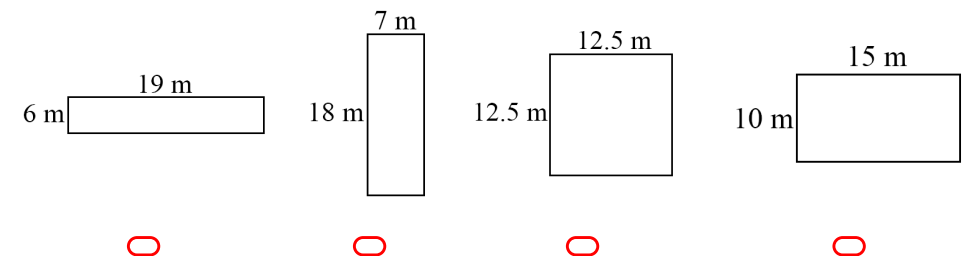
What is the perimeter of the shape?

10 cm 14 cm 12 cm 20 cm
☐ ☐ ☐ ☐

3. Measurement, NAP-32095

A resort has 4 pools.

Which pool has the largest surface area?



4. Measurement, NAP-32094



What is the best estimate for the mass of this basketball?

- ☐ 20 grams
 ☐ 200 grams
 ☐ 20 kilograms
 ☐ 200 kilograms

5. Number, NAP-26177

A bullet fired from a hand gun travels at 390 metres per second.

What is the speed of the bullet in metres per minute?

- ☐ 39 000
 ☐ 23 400
 ☐ 6.5
 ☐ 3.9

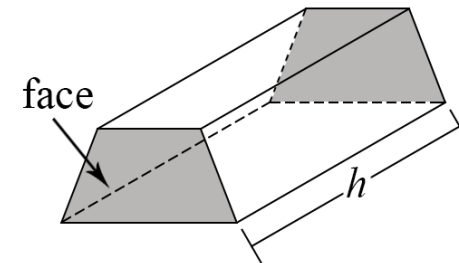
6. Measurement, NAP-55611

Which of these can be measured in grams?

- ☐ length
 ☐ volume
 ☐ mass
 ☐ area

7. Algebra, NAP-79137

The surface area (S) of the prism below can be calculated by multiplying the perimeter of its face (p) by its height (h), and adding twice the area of its face (F).

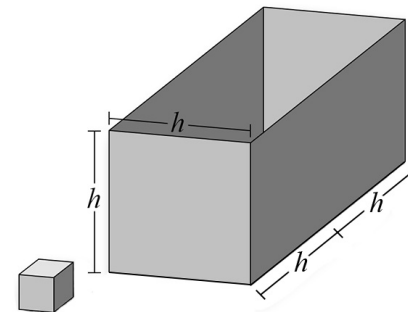


Which one of these formulas could be used for this calculation?

- ☐ $S = 2phF$
☐ $S = ph + F$
☐ $S = ph + 2F$
☐ $S = 2ph + 2F$

8. Measurement, NAP-73221

A shoebox is square at each end and twice as long as it is high, as shown in the diagram.

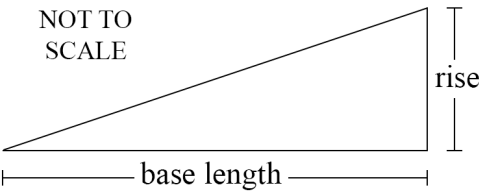


Which of these is the closest estimate for the number of cubes that would fill the box?

- ☐ 16
 ☐ 60
 ☐ 130
 ☐ 500

9. Geometry, NAP-32121

A ramp rises 1 metre for every 8 metres of its base length.

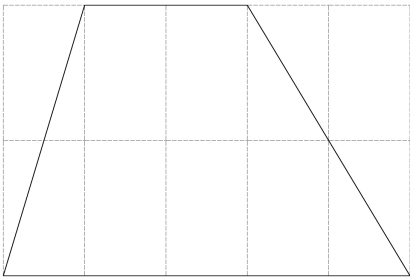


If it rises 0.7 metres, what is its base length?

- 5.6 metres
☐
- 8.7 metres
☐
- 11.4 metres
☐
- 16 metres
☐

10. Measurement, NAP-49701

A trapezium is constructed on a grid of 10 rectangles.
Each rectangle measures 3 cm × 5 cm.

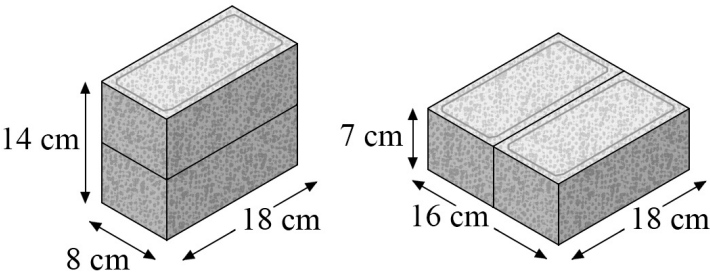


What is the area of the trapezium?

- 85 cm²
☐
- 95 cm²
☐
- 105 cm²
☐
- 125 cm²
☐

11. Measurement, NAP-38036

Two bricks can be joined to make three different rectangular prisms. Two of them are shown here.



What would be the measurements of the third prism?

- ☐ 18 cm by 16 cm by 7 cm
- ☐ 36 cm by 8 cm by 7 cm
- ☐ 32 cm by 18 cm by 7 cm
- ☐ 36 cm by 14 cm by 8 cm

12. Measurement, NAP-61561

Pillar draws a circle with a diameter of 3.2 cm.
What is the circumference of the circle to the nearest centimetre?

- 5 cm
☐
- 6 cm
☐
- 10 cm
☐
- 20 cm
☐

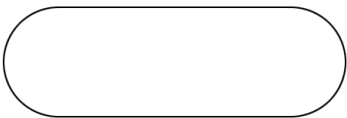
13. Measurement, NAP-08601

Grant cut two semicircles from a rectangle to create Shape 1.



Shape 1

He then joins the semicircles to each end of an identical rectangle to create Shape 2.



Shape 2

Which of the following statements is true about Shape 1 and Shape 2?

- ☐ They have the same area and the same perimeter.
- ☐ They have different areas and the same perimeter.
- ☐ They have the same area and different perimeters.
- ☐ They have different areas and different perimeters.

14. Measurement, NAP-96772

Cecil is designing a rectangular garden bed for growing vegetables in his yard.

He wants the garden bed to be at least 60 square metres.

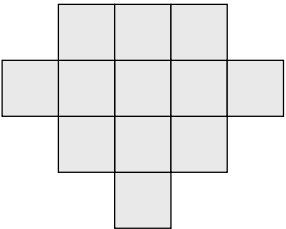
He has dug out two sides, each 12.7 metres in length.

What is the smallest possible length of each of the other two sides, rounded to one decimal place?

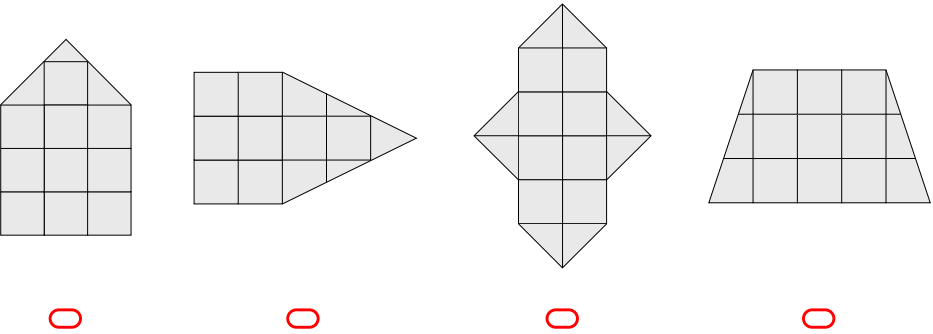
- 4 metres** **4.7 metres** **4.8 metres** **5 metres**
- ☐ ☐ ☐ ☐

15. Measurement, NAP-72923

Cassy drew this shape on some grid paper.

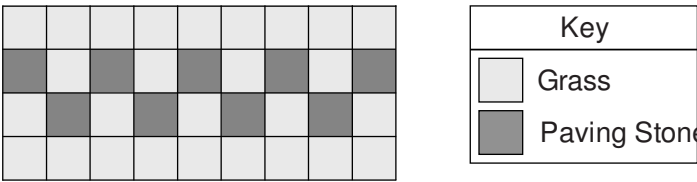


Select the two shapes with the same area as Cassy's shape.



16. Measurement, NAP-08605

Here is a plan of Dave's backyard.



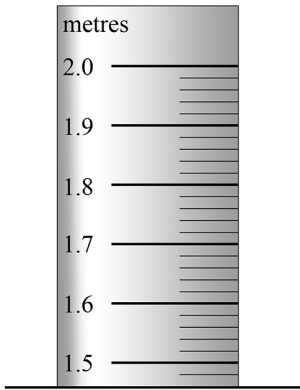
The total area of the paving stones is 18 m².

What is the total area of the grass in Dave's backyard?

- 36 m²** **48 m²** **54 m²** **72 m²**
- ☐ ☐ ☐ ☐

17. Measurement, NAP-26209

A flood level indicator measures the height of a river.



What is the height of the river?

- 1.44 metres** **1.46 metres** **1.48 metres** **1.52 metres** **1.54 metres**
- ☐ ☐ ☐ ☐ ☐

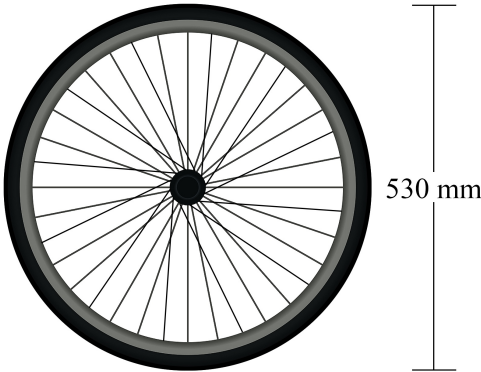
18. Measurement, NAP-02688

A child's bike tyre has a circumference of 90 cm.
Which of these is closest to the radius of the circle?

- 5.4 cm** **14.3 cm** **28.6 cm** **141.4 cm**
- ☐ ☐ ☐ ☐

19. Measurement, NAP-96773

A wheel on a bicycle has a diameter of 530 mm.

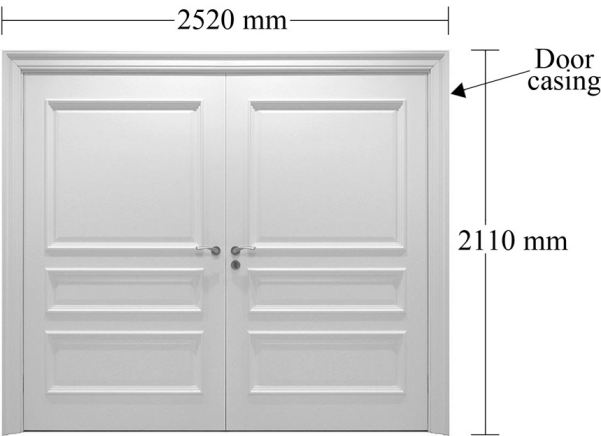


The wheel turns 12 times.
Approximately how many metres has the bicycle travelled?

- 6 m** **20 m** **40 m** **399 m**
- ☐ ☐ ☐ ☐

20. Measurement, NAP-26237

Guy is placing door casing on the top and both sides of 2 doors, as pictured below.

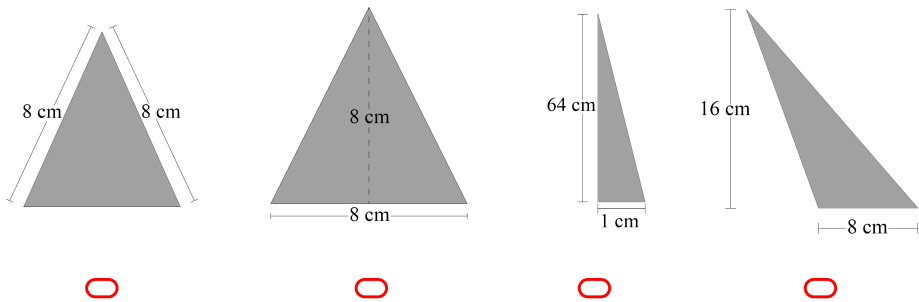


How many metres of door casing does he need?

- 4.63 m
- 6.74 m
- 9.26 m
- 67.4 m
- ☐
- ☐
- ☐
- ☐

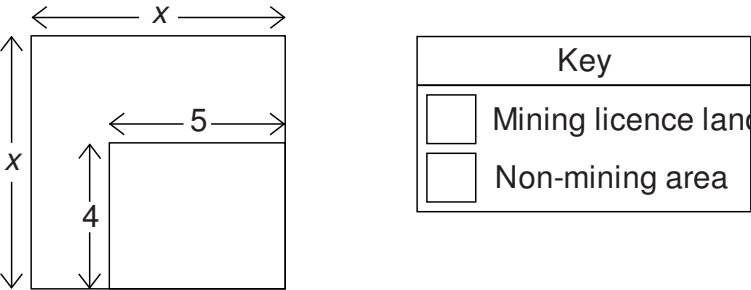
21. Measurement, NAP-96775

Which one of the following triangles has an area of 64 cm² ?
(Triangles are not drawn to scale)



22. Algebra, NAP-08636

The plan of a square area of land is drawn below.
A mining licence is granted over part of the land.
All measurements are given in kilometres.



The area of land the mining licence covers is

- $x^2 - 20$
- $x^2 + 20$
- $2x^2 - 9$
- $2x^2 - 20$
- ☐
- ☐
- ☐
- ☐

23. Measurement, NAP-38098

A rectangle has an area of 1 square metre.
It has a length of 1222 millimetres.
What is its width to the nearest millimetre?

 millimetres